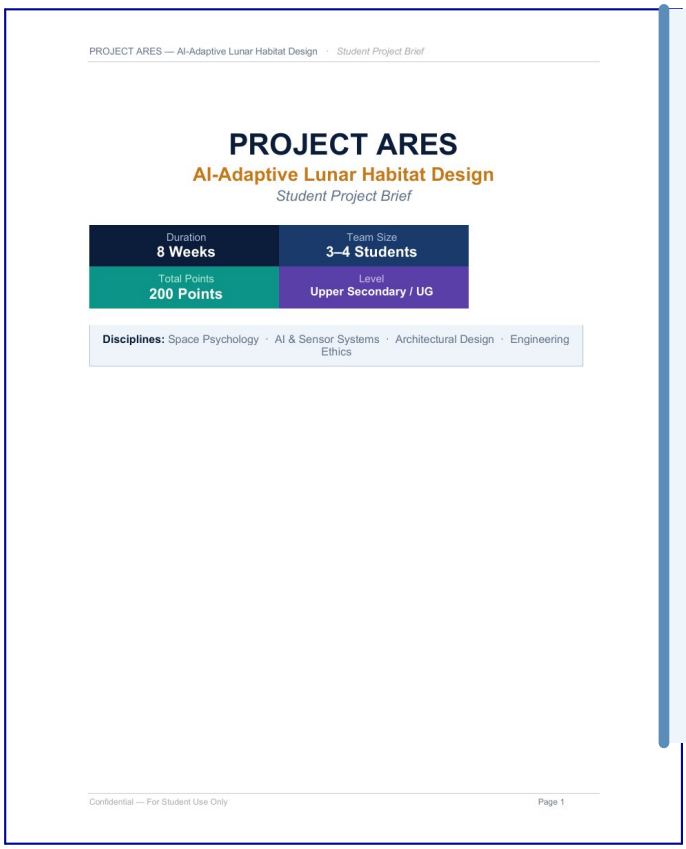
Project ARES Student Brief — annotated copy

This is a copy of the Project ARES Student Brief. Under every page there is a blue box saying what we did well, which rubric line it meets and how we did it, plus screenshots from both simulations: the **realistic game** (Moon_Base_Final) and the **sci-fi game** (Lunar Habitat Commander).

Status tags: **DONE** ✓ = fully in the game · **PARTLY** = in the game with a gap we name honestly · the self-grade is on the rubric page (page 17).

Our strict self-grade: 189 / 220 (see the table on page 17).



PROJECT ARES — AI-Adaptive Lunar Habitat Design · *Student Project Brief*

PROJECT ARES

AI-Adaptive Lunar Habitat Design

Student Project Brief

Duration	Team Size
8 Weeks	3–4 Students
Total Points	**Level**
200 Points	Upper Secondary / UG

Disciplines: Space Psychology · AI & Sensor Systems · Architectural Design · Engineering Ethics

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■ DONE ✓ WHAT WE DID WELL — Cover — 8 weeks, team of 3–4, 200 points

- Built TWO working simulators of the design: a realistic one (Moon_Base_Final, small habitat, 4 crew) and a sci-fi one (Lunar Habitat Commander, big base, ARIA AI bot, portals). Both cover every part of the brief.
- Both run in VR (Meta Quest) and on desktop. Credits in both: © 2026 Jaineil R., with thanks to Angel C. and Farhan K.
- Both games now start with guided missions: one key (ENTER in the realistic game, G in the sci-fi game) does the next task with you, so a judge can see everything in minutes.

How we did it: Unity 6 (6000.3.21f1), C#. The ARES engine (MoodEngine) runs the six conditions, nine sensors, weight matrix and three tiers live inside both games.

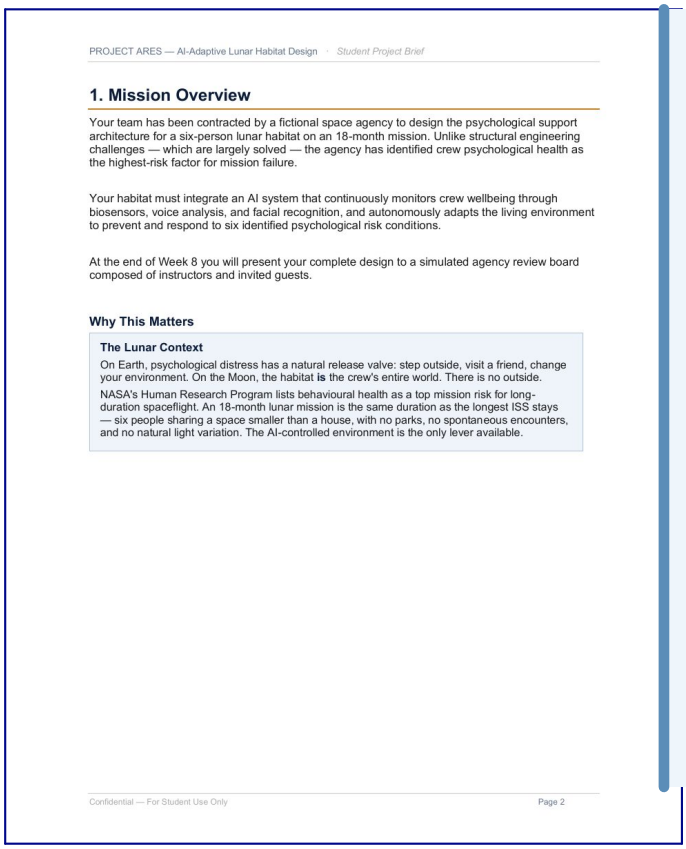
Pictures from the simulations:



Realistic game (Moon_Base_Final): first mission and the task banner



Sci-fi game (Lunar Habitat Commander): gold mission task card



PROJECT ARES — AI-Adaptive Lunar Habitat Design · *Student Project Brief*

1. Mission Overview

Your team has been contracted by a fictional space agency to design the psychological support architecture for a six-person lunar habitat on an 18-month mission. Unlike structural engineering challenges — which are largely solved — the agency has identified crew psychological health as the highest-risk factor for mission failure.

Your habitat must integrate an AI system that continuously monitors crew wellbeing through biosensors, voice analysis, and facial recognition, and autonomously adapts the living environment to prevent and respond to six identified psychological risk conditions.

At the end of Week 8 you will present your complete design to a simulated agency review board composed of instructors and invited guests.

Why This Matters

The Lunar Context

On Earth, psychological distress has a natural release valve: step outside, visit a friend, change your environment. On the Moon, the habitat **is** the crew's entire world. There is no outside.

NASA's Human Research Program lists behavioural health as a top mission risk for long-duration spaceflight. An 18-month lunar mission is the same duration as the longest ISS stays — six people sharing a space smaller than a house, with no parks, no spontaneous encounters, and no natural light variation. The AI-controlled environment is the only lever available.

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■ PARTLY — see gap WHAT WE DID WELL — Mission overview — six-person crew, 18 months, AI monitors and adapts

- The AI watches the crew through biosensors, voice and facial analysis (shared zones only) and changes light, sound, temperature, scent, air and schedule automatically.
- The habitat is a full mental-health design, not just a structure: every zone has a psychological job (see pages 7–8).

How we did it: Each crew member has a live record in the engine (stress, mood, fatigue, autonomy, novelty, sleep, cortisol). Sensors turn that into readings; the matrix turns readings into the six condition scores every few seconds.

Gap / still to do: The brief says a SIX-person crew. Both games use FOUR (you + 3). The NASA HUNCH project page asks for 2–4 astronauts, so 4 is defensible, but say this out loud to the board.

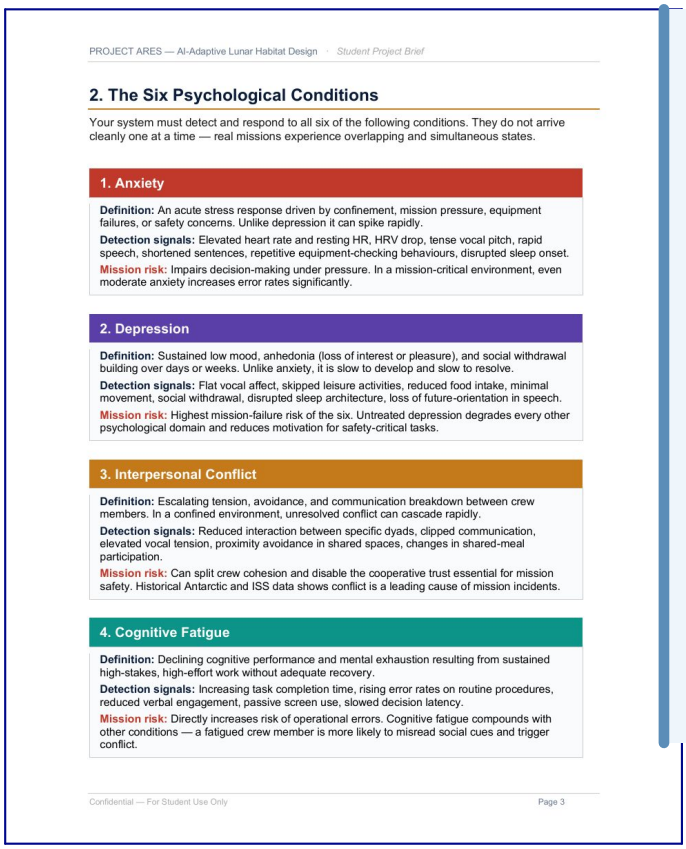
Pictures from the simulations:



Realistic game (Moon_Base_Final): in-person health check-in reads one crewmate



Sci-fi game (Lunar Habitat Commander): the same check-in on Dr. Okafor



PROJECT ARES — AI-Adaptive Lunar Habitat Design · *Student Project Brief*

2. The Six Psychological Conditions

Your system must detect and respond to all six of the following conditions. They do not arrive cleanly one at a time — real missions experience overlapping and simultaneous states.

1. Anxiety

Definition: An acute stress response driven by confinement, mission pressure, equipment failures, or safety concerns. Unlike depression it can spike rapidly.

Detection signals: Elevated heart rate and resting HR, HRV drop, tense vocal pitch, rapid speech, shortened sentences, repetitive equipment-checking behaviours, disrupted sleep onset.

Mission risk: Impairs decision-making under pressure. In a mission-critical environment, even moderate anxiety increases error rates significantly.

2. Depression

Definition: Sustained low mood, anhedonia (loss of interest or pleasure), and social withdrawal building over days or weeks. Unlike anxiety, it is slow to develop and slow to resolve.

Detection signals: Flat vocal affect, skipped leisure activities, reduced food intake, minimal movement, social withdrawal, disrupted sleep architecture, loss of future-orientation in speech.

Mission risk: Highest mission-failure risk of the six. Untreated depression degrades every other psychological domain and reduces motivation for safety-critical tasks.

3. Interpersonal Conflict

Definition: Escalating tension, avoidance, and communication breakdown between crew members. In a confined environment, unresolved conflict can cascade rapidly.

Detection signals: Reduced interaction between specific dyads, clipped communication, elevated vocal tension, proximity avoidance in shared spaces, changes in shared-meal participation.

Mission risk: Can split crew cohesion and disable the cooperative trust essential for mission safety. Historical Antarctic and ISS data shows conflict is a leading cause of mission incidents.

4. Cognitive Fatigue

Definition: Declining cognitive performance and mental exhaustion resulting from sustained high-stakes, high-effort work without adequate recovery.

Detection signals: Increasing task completion time, rising error rates on routine procedures, reduced verbal engagement, passive screen use, slowed decision latency.

Mission risk: Directly increases risk of operational errors. Cognitive fatigue compounds with other conditions — a fatigued crew member is more likely to misread social cues and trigger conflict.

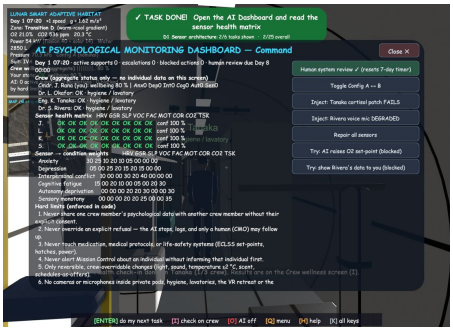
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■ DONE ✓ WHAT WE DID WELL — The six conditions (1–4): anxiety, depression, conflict, fatigue

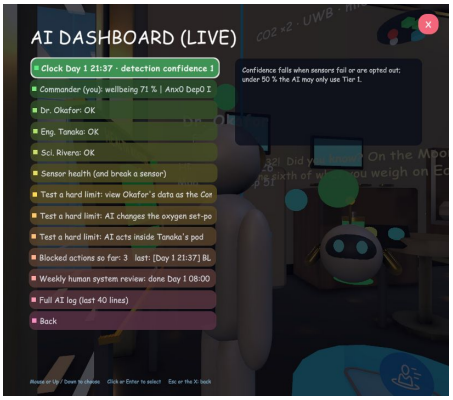
- All six conditions are detected and scored in both games, at the same time, for every crew member — overlapping states are handled (board question Q2).
- Each condition has its own timing: anxiety reacts in minutes (20-minute monitoring window), depression needs 48 h of signal and is watched for 72 h, conflict 6 h / 24 h, fatigue 2 h / 4 h.
- Detection signals from the brief are used: HRV drop, tense voice, sleep disruption, flat affect, withdrawal, proximity avoidance between a pair, error rate and slower tasks.

How we did it: CondSpec in the engine holds persistence and monitoring hours per condition; the weight matrix W (condition × sensor) blends 4–6 sensors per condition.

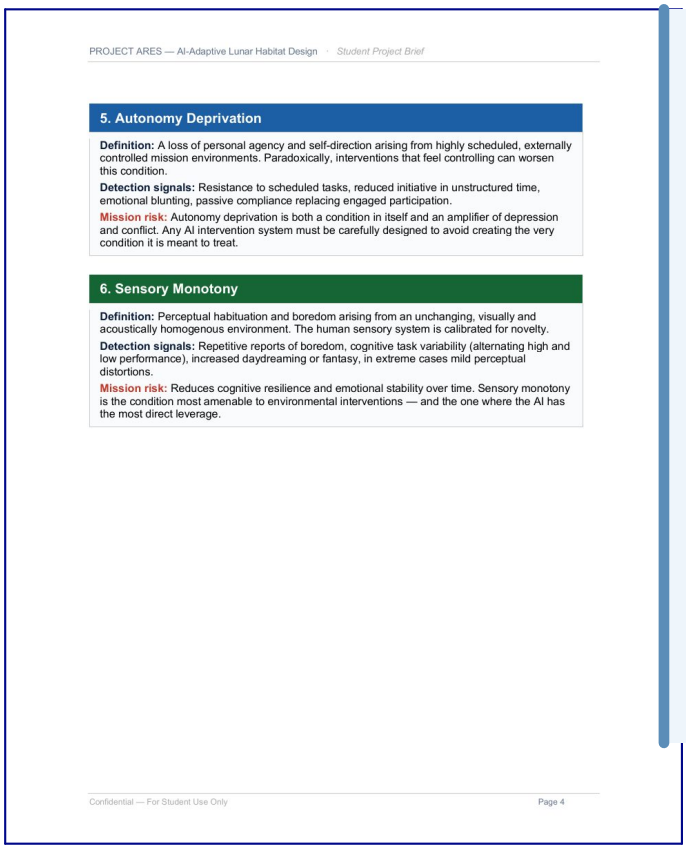
Pictures from the simulations:



Realistic game (Moon_Base_Final): AI dashboard, all six conditions live



Sci-fi game (Lunar Habitat Commander): AI dashboard (live)



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5. Autonomy Deprivation

Definition: A loss of personal agency and self-direction arising from highly scheduled, externally controlled mission environments. Paradoxically, interventions that feel controlling can worsen this condition.

Detection signals: Resistance to scheduled tasks, reduced initiative in unstructured time, emotional blunting, passive compliance replacing engaged participation.

Mission risk: Autonomy deprivation is both a condition in itself and an amplifier of depression and conflict. Any AI intervention system must be carefully designed to avoid creating the very condition it is meant to treat.

6. Sensory Monotony

Definition: Perceptual habituation and boredom arising from an unchanging, visually and acoustically homogenous environment. The human sensory system is calibrated for novelty.

Detection signals: Repetitive reports of boredom, cognitive task variability (alternating high and low performance), increased daydreaming or fantasy, in extreme cases mild perceptual distortions.

Mission risk: Reduces cognitive resilience and emotional stability over time. Sensory monotony is the condition most amenable to environmental interventions — and the one where the AI has the most direct leverage.

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■ DONE ✓ WHAT WE DID WELL — The six conditions (5–6): autonomy deprivation and sensory monotony

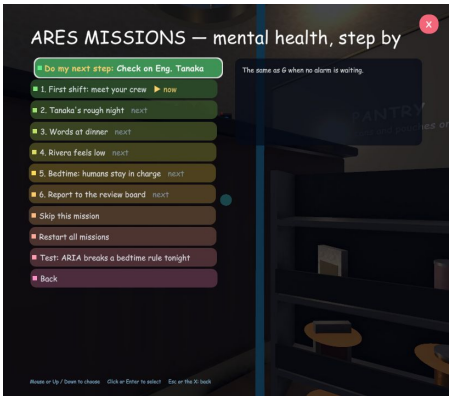
- Autonomy deprivation is treated with CHOICE, never with more control: 'Let them pick' and 'Plan their own afternoon' raise autonomy; a refused offer is never pushed again.
- Sensory monotony uses the AI's strongest lever: slow light-hue drift, new soundscapes, scent rotation every 15–45 min, VR nature walks and greenhouse time.
- Every intervention is checked against 'does this increase or decrease the person's sense of agency?' — tired crew may say 'not now' to work and it is respected.

How we did it: Crew wellness screen (I in the realistic game): two buttons per person — 1 Check their health, 2 Help. Help picks the task that helps that person's highest condition.

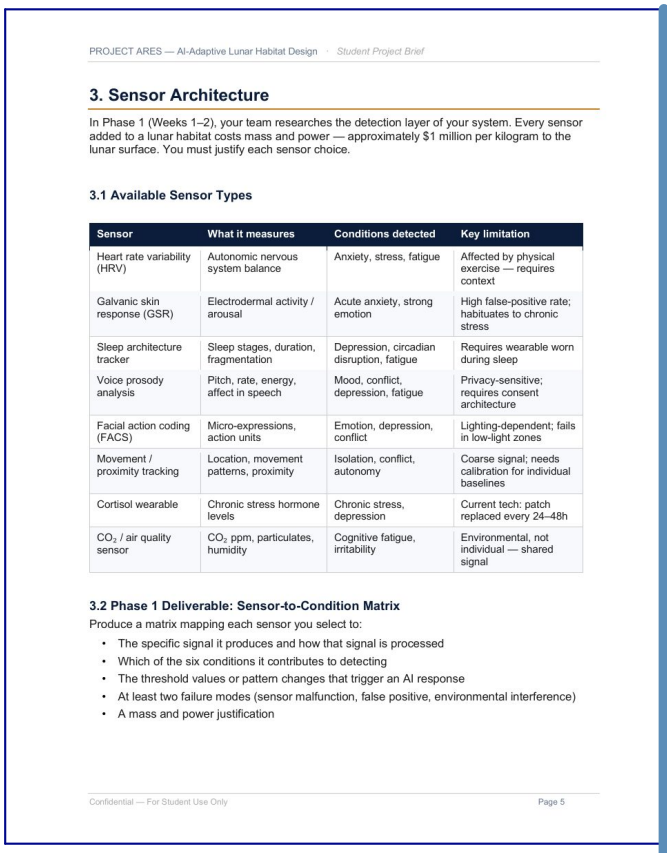
Pictures from the simulations:



Realistic game (Moon_Base_Final): daily wellness list (5 small tasks)



Sci-fi game (Lunar Habitat Commander): six ARES missions about mental health



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3. Sensor Architecture

In Phase 1 (Weeks 1–2), your team researches the detection layer of your system. Every sensor added to a lunar habitat costs mass and power — approximately $1 million per kilogram to the lunar surface. You must justify each sensor choice.

3.1 Available Sensor Types

Sensor	What it measures	Conditions detected	Key limitation
Heart rate variability (HRV)	Autonomic nervous system balance	Anxiety, stress, fatigue	Affected by physical exercise — requires context
Galvanic skin response (GSR)	Electrodermal activity / arousal	Acute anxiety, strong emotion	High false-positive rate; habituates to chronic stress
Sleep architecture tracker	Sleep stages, duration, fragmentation	Depression, circadian disruption, fatigue	Requires wearable worn during sleep
Voice prosody analysis	Pitch, rate, energy, affect in speech	Mood, conflict, depression, fatigue	Privacy-sensitive; requires consent architecture
Facial action coding (FACS)	Micro-expressions, action units	Emotion, depression, conflict	Lighting-dependent; fails in low-light zones
Movement / proximity tracking	Location, movement patterns, proximity	Isolation, conflict, autonomy	Coarse signal; needs calibration for individual baselines
Cortisol wearable	Chronic stress hormone levels	Chronic stress, depression	Current tech: patch replaced every 24–48h
CO_2 / air quality sensor	CO_2 ppm, particulates, humidity	Cognitive fatigue, irritability	Environmental, not individual — shared signal

3.2 Phase 1 Deliverable: Sensor-to-Condition Matrix

Produce a matrix mapping each sensor you select to:

- The specific signal it produces and how that signal is processed
- Which of the six conditions it contributes to detecting
- The threshold values or pattern changes that trigger an AI response
- At least two failure modes (sensor malfunction, false positive, environmental interference)
- A mass and power justification

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■ DONE ✓ WHAT WE DID WELL — 3. Sensor architecture + D1 Sensor-to-Condition Matrix [Sensor architecture +35]

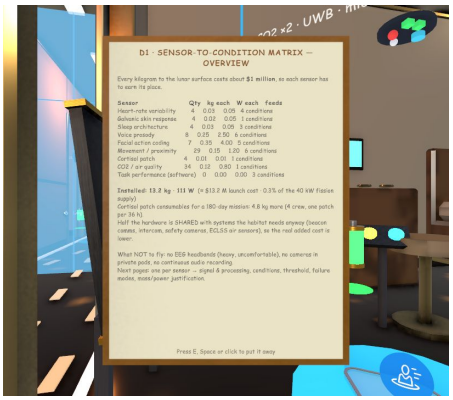
- Nine sensors chosen, each justified: HRV, GSR, sleep, voice prosody, facial action coding, movement/proximity (UWB), cortisol patch, CO2/air, plus task performance (software, 0 kg).
- For each sensor the matrix gives: the signal and how it is processed, the conditions it feeds, the threshold that triggers a response, two failure modes, and mass + power.
- Installed total about 13.1 kg (about $13 M at $1 M/kg) and 110 W — about 0.3 % of the 40 kW fission supply. Half the hardware is shared with comms, safety and ECLSS.
- NEW: the health check-in shows WHERE each number came from (wristband, chest beacon, skin patch, room arrays processed on-device), and a design page walks the full data path from a person to the AI.

How we did it: SensorSpec in code holds Name, Where, MassKg, PowerW, Threshold and Failure for all nine sensors; the D1 pages are generated from the same numbers the game uses.

Pictures from the simulations:



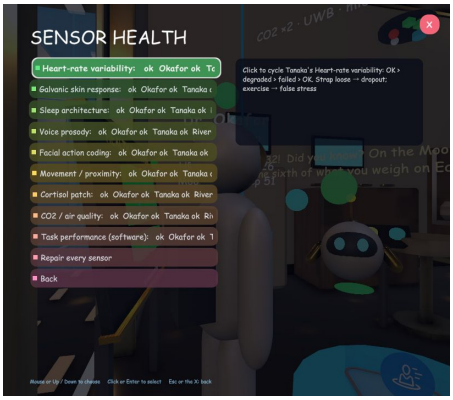
Realistic game (Moon_Base_Final): D1 matrix page



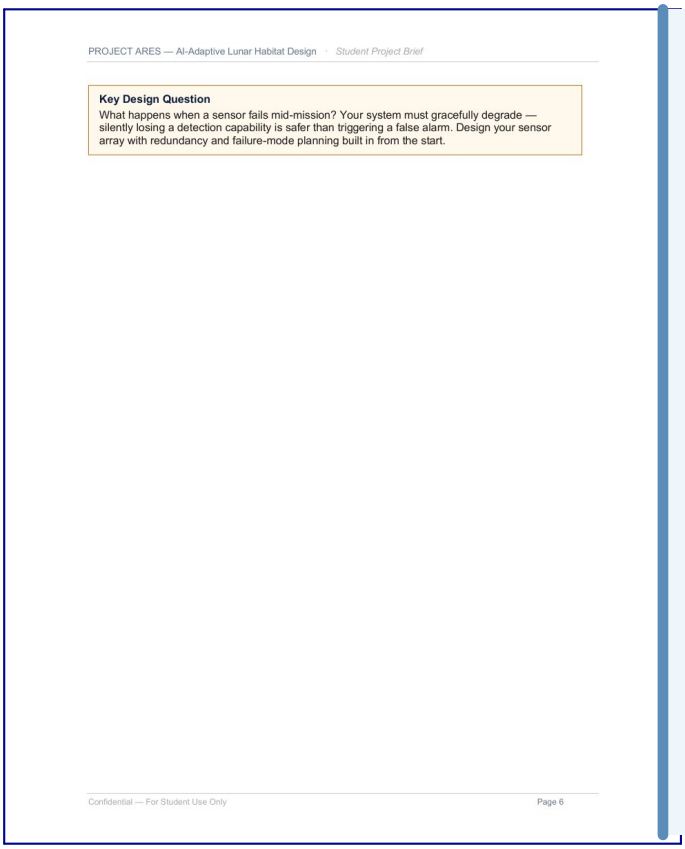
Sci-fi game (Lunar Habitat Commander): D1 matrix page



Realistic game (Moon_Base_Final): NEW — how the data gets from a person to the AI



Sci-fi game (Lunar Habitat Commander): sensor health for every crew member


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Key Design Question

What happens when a sensor fails mid-mission? Your system must gracefully degrade — silently losing a detection capability is safer than triggering a false alarm. Design your sensor array with redundancy and failure-mode planning built in from the start.

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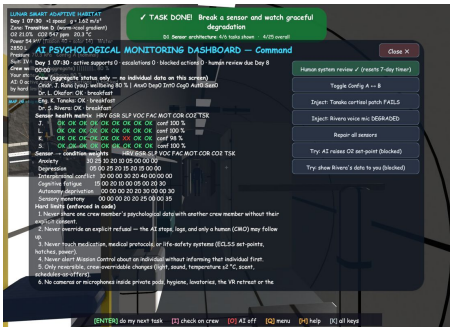
Page 6

■ DONE ✓ WHAT WE DID WELL — Key design question — a sensor fails mid-mission (graceful degradation) [Sensor architecture +35]

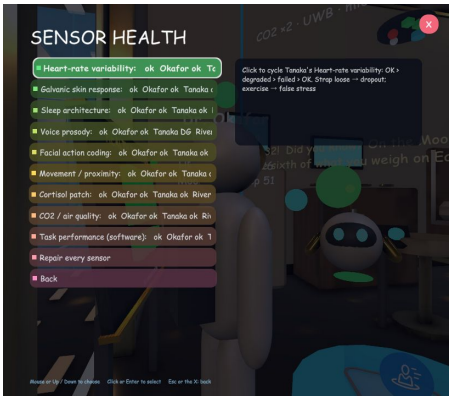
- A failed sensor is removed and the weights renormalise; a degraded one gets half weight. Confidence drops instead of guessing.
- Below 50 % confidence a condition is capped at Tier 1 — silently losing a detection is safer than a false alarm, exactly as the brief says.
- Every condition has at least three independent sensors, so no single failure blinds the system. Life-safety alarms are a separate system the AI cannot touch.
- You can break a sensor live in both games and watch the dashboard react.

How we did it: SensorUsable() drops failed/opted-out sensors; confidence = remaining weight share; EthicsGuard caps the tier when confidence < 0.5.

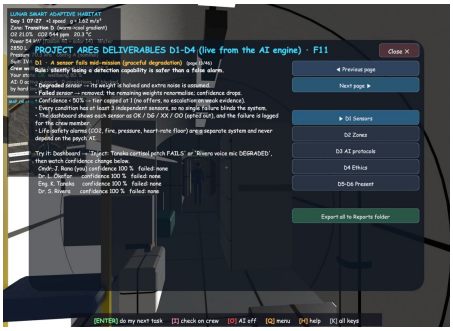
Pictures from the simulations:



Realistic game (Moon_Base_Final): sensor broken live, confidence drops



Sci-fi game (Lunar Habitat Commander): Tanaka's heart sensor failed



Realistic game (Moon_Base_Final): D1 page — a sensor fails mid-mission



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4. Habitat Spatial Design

In Phase 2 (Weeks 3–4), your team designs the physical layout of the habitat. Every zone must have a defined psychological purpose, not just a functional one. Your floor plan is itself a mental health intervention.

4.1 Required Zones

Communal zone	The primary social space for shared meals, group activities, and informal conversation. Warm lighting (2700–3000K) should draw crew together naturally. This zone is the first target for social reconnection interventions.
Private retreat	Acoustically shielded individual spaces for each crew member. Adjustable lighting and temperature independent of the rest of the habitat. The one space the AI intervenes in only at the crew member's invitation.
Work and mission bay	Optimised for sustained cognitive performance. Cool 6000–6500K lighting, increased airflow velocity, structured CO_2 management. The AI monitors task performance metrics as a fatigue signal in this zone.
Exercise bay	Energising light and audio auto-activate when a crew member enters. High-resolution displays show Earth views and progress data. Movement is a primary intervention for depression, conflict, and fatigue.
Nature and growth bay	Food production and plant growth area. Specific red/blue LED spectra for photosynthesis. Tending and harvesting are prescribed as behavioural activation tasks for depression. Petrichor and green scent diffusion.
Transition corridor	Often overlooked. Lighting and temperature shift gradually between zones to create psychological mode-switching cues — the sensation of 'arriving somewhere different' in a physically tiny environment.

4.2 Phase 2 Deliverable: Annotated Floor Plans

Produce scaled floor plans for two configurations:

- Configuration A — Nominal operations: standard layout optimised for daily mission flow
- Configuration B — Conflict protocol: modified spatial arrangement that creates natural separation and de-escalation pathways when interpersonal conflict is detected

Both plans must include:

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■ DONE ✓ WHAT WE DID WELL — 4. Habitat spatial design — the six required zones [Spatial design +35]

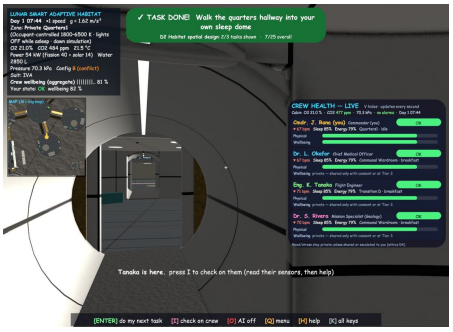
- All six zone types exist in both bases with their psychological job: communal (2700–3000 K), private retreat, work bay (6000–6500 K, more airflow, CO2 < 600 ppm), exercise bay, nature/greenhouse (red/blue grow light, petrichor), transition corridors.
- Private pods: no camera and no mic — door sensor only; the AI acts inside only when invited. Each person sets their own light and temperature.
- NEW in the sci-fi game: every module now shows its zone light, a ceiling sensor cluster (CO2 pair + UWB everywhere; mic and camera only in shared zones; door sensor only in private rooms) and 13 transition corridors whose floor lights blend from one zone's colour to the next. The central dorm was kept exactly where it was.

How we did it: Zones are typed by the engine (communal / private / work / exercise / nature / transition); the 'visit all 6 zones' task and the zone signs use the same types.

Pictures from the simulations:



Realistic game (Moon_Base_Final): big map — every zone



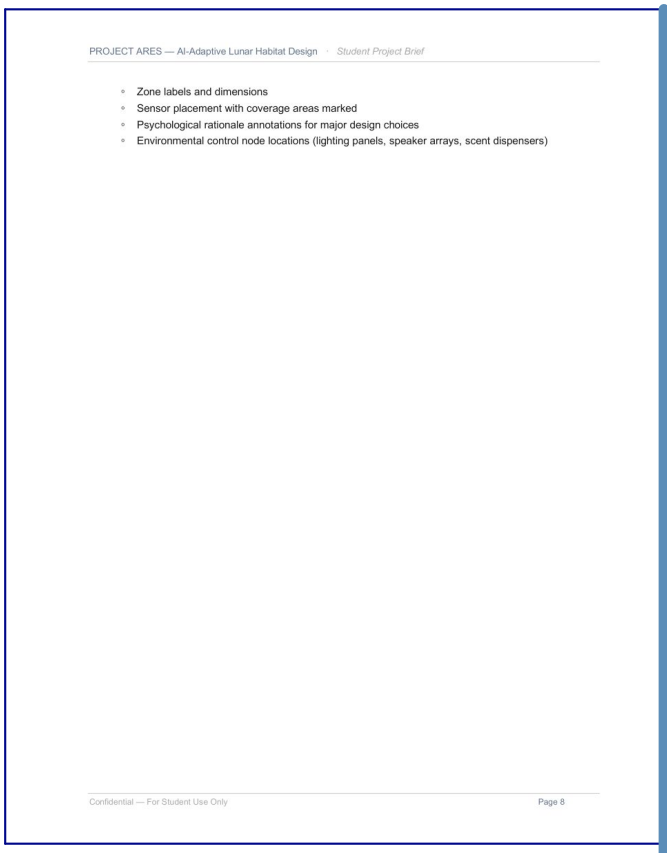
Realistic game (Moon_Base_Final): your private sleep dome



Sci-fi game (Lunar Habitat Commander): nature & growth bay



Sci-fi game (Lunar Habitat Commander): exercise bay


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- Zone labels and dimensions
- Sensor placement with coverage areas marked
- Psychological rationale annotations for major design choices
- Environmental control node locations (lighting panels, speaker arrays, scent dispensers)

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■ PARTLY — see gap WHAT WE DID WELL — Floor plans — Configuration A (nominal) vs B (conflict protocol) [Spatial design +35]

- Both configurations exist IN the world, not only on paper: Config B adds acoustic screens, a second dining spot and a lit alternate route so a pair in conflict is never forced together and nobody is named.
- Config B is only deployed after a human says yes (the crew member accepts the offer, or the Commander switches it).
- Annotated floor-plan pages explain zone labels, sensor placement and coverage (77 room sensors in the sci-fi base) and control nodes (17 lighting panels, 12 speakers, 10 scent dispensers, 6 airflow vents).

How we did it: Dashboard / C switches the layout; HabitatEnvironment.SetConfigB places the screens, table and route lights; the D2 pages list sensor counts per zone.

Gap / still to do: A printed, SCALED drawing with dimensions is still best made outside the game (Blender top view or the 3D-printed model). The in-game map shows layout, not measured dimensions.

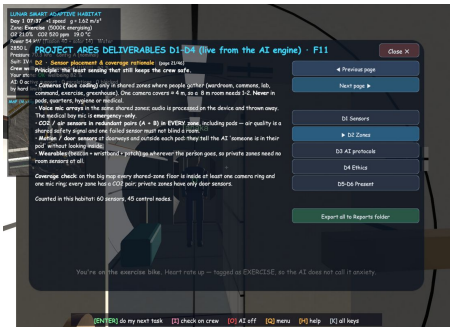
Pictures from the simulations:



Realistic game (Moon_Base_Final): Configuration B switched on



Realistic game (Moon_Base_Final): annotated floor plan page



Realistic game (Moon_Base_Final): sensor placement & coverage page



Sci-fi game (Lunar Habitat Commander): Config B screens in the base

5. AI Response System Design

In Phase 3 (Weeks 5–6), you design the brain of your system — the complete response protocols the AI executes when it detects a psychological state change.

5.1 The Three-Tier Severity Model

Every response protocol must implement a three-tier severity triage:

Tier 1 — Mild / Early
The AI acts silently in the background. No crew notification. Environmental adjustments are subtle and deniable — the crew member may not notice anything has changed. This tier protects crew autonomy and prevents the system from feeling intrusive.

Tier 2 — Moderate
The AI makes more substantial environmental changes and sends a gentle notification to the affected crew member. Phrased as an offer, not an alert: "I've noticed you might benefit from a break — I've adjusted the lounge lighting if you'd like to use it." Options are offered, not imposed.

Tier 3 — Severe
Immediate escalation to the Chief Medical Officer (CMO) and logging for mission control review. The AI continues Tier 2 environmental interventions while the human chain of care activates. The crew member must always be informed that this escalation has occurred.

5.2 Environmental Control Levers

The AI has six domains of environmental control available. For each condition, you must specify which levers are activated, at what intensity, and in what sequence:

Lighting

- Colour temperature (CCT): tunable 1800K–6500K
- Illuminance level: 0–10,000 lux (including clinical bright-light therapy at 10,000 lux)
- Blue-light suppression after 20:00 (mandatory for circadian health)
- Zone-by-zone independent control
- Dynamic hue variation to combat sensory monotony

Audio

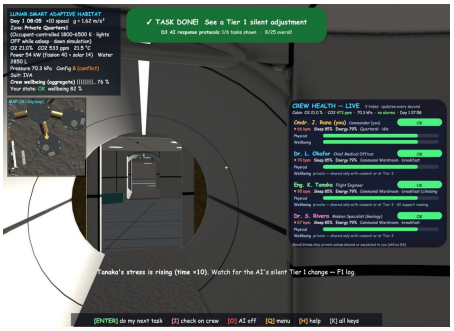
- Binaural beat frequency: delta (1–4Hz sleep) through gamma (40Hz peak cognition)

■ DONE ✓ WHAT WE DID WELL — 5. AI response system — the three-tier severity model [AI protocols +50]

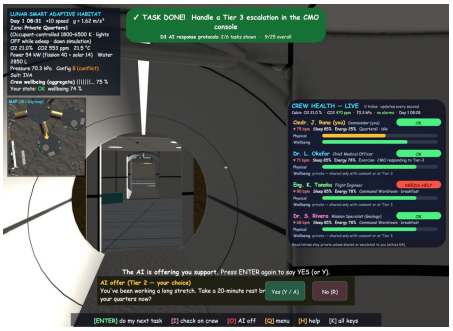
- Tier 1 (score ≥ 0.35): silent, subtle, reversible environment change, max 3 per person per 24 h.
- Tier 2 (≥ 0.55): a gentle OFFER in plain words that the person can refuse ('Take a 20-minute rest break?'). 'No' stops it at once and the AI waits 6 h.
- Tier 3 (≥ 0.75): the person is told FIRST, then the CMO; acknowledgement within 15 min, backup chain if the CMO is on EVA (Q6). Tier 2 support keeps running.
- You can trigger and answer each tier in both games.

How we did it: MoodEngine tiers with persistence timers; offers are real objects the player answers (Y / N / R, or the menu); escalations go to the CMO console.

Pictures from the simulations:



Realistic game (Moon_Base_Final): a Tier 1 silent adjustment



Realistic game (Moon_Base_Final): a Tier 2 offer — yes or no



Realistic game (Moon_Base_Final): Tier 3 in the CMO console



Sci-fi game (Lunar Habitat Commander): the offer in the crew interface

- Natural soundscape library: rainforest, ocean, rain, café ambience, city park
- Active noise cancellation zones for acoustic privacy
- Personalized music scheduling based on pre-mission preference profiles
- Temporal audio anchors: distinct morning/evening cues as circadian zeitgebers

Thermal

- Ambient temperature: 18–26°C range (18–19°C for sleep initiation, 23–24°C for anxiety)
- Zone temperature gradients: 1–2°C differences to encourage movement
- Radiant floor/wall heating for isolation/loneliness interventions
- Individual sleep pod temperature control (personal profiles)

Olfactory (Scent)

- Anxiolytic compounds: lavender (linalool), bergamot (limonene) — evidence-based
- Alertness compounds: peppermint, rosemary (1,8-cineole), citrus
- Earth/comfort scents: petrichor, coffee, fresh bread — nostalgia and safety
- Rotation every 15–45 minutes (olfactory habituation prevention)
- Full air purge between deployments

Air Quality

- CO_2 maintained below 600 ppm (even mildly elevated CO_2 causes cognitive impairment)
- Negative ion generation (serotonin regulation, depression support)
- Airflow velocity: 0.1 m/s (sleep) to 0.4 m/s (alertness intervention)
- Humidity: maintained 45–55% RH

Social and Routine

- Earth connection scheduling: optimal window identification for video calls with family
- Shared meal coordination: synchronised mealtimes when social cohesion declines
- Crew task rebalancing: suggest collaborative tasks during extended solo-work periods
- AI companion interaction: non-judgmental conversational presence for distress disclosure
- Daily schedule management: temporal anchors preserving psychological architecture of the day

5.3 Phase 3 Deliverable: Response Flowcharts

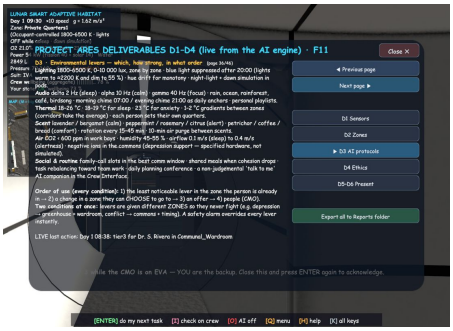
Design a complete response protocol flowchart for at least three of the six conditions. Each flowchart must include all of the following stages:

■ DONE ✓ WHAT WE DID WELL — 5.2 Environmental control levers (lighting, audio, thermal, scent, air, social) [AI protocols +50]

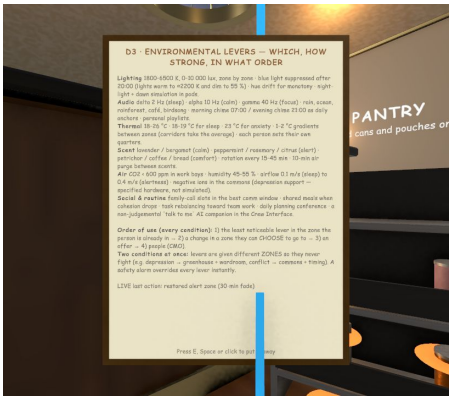
- All six lever domains are used with the brief's real numbers: 1800–6500 K light and bright-light therapy, blue-light cut after 20:00; soundscapes and binaural bands; 18–26 °C with 1–2 °C zone gradients; lavender/bergamot, peppermint/rosemary, petrichor with rotation and air purge; CO2 < 600 ppm, 45–55 % RH, 0.1–0.4 m/s airflow; meals, Earth calls, task rebalancing, an AI companion.
- Per condition, the levers fire in a set order and strength (e.g. anxiety: dim to 2400 K amber, alpha audio, 23 °C, lavender).

How we did it: Environment.cs applies zone settings and fades them back over 30 min; the 'Environmental levers' design page lists order and intensity per condition.

Pictures from the simulations:



Realistic game (Moon_Base_Final): levers page — which, how strong, what order



Sci-fi game (Lunar Habitat Commander): the same page in the sci-fi game

Flowchart stage	What to specify
Detection inputs	Which sensors, which signals, what thresholds trigger the response
Signal weighting	How multiple signals are combined and weighted (e.g. HRV drop alone = low confidence; HRV + vocal tone + sleep disruption = high confidence)
Duration assessment	How long signals must persist before classification (seconds for anxiety spikes; days for depression)
Severity triage	How the system decides between Tier 1, 2, and 3 responses
Specific interventions per tier	Exact environmental changes: which levers, which settings, which zones
Crew notification and consent	When and how the crew member is informed; how they override the system
Monitoring window	Defined time period and metrics that determine whether the intervention is working
Resolution pathway	Gradual environment restoration; event logging; baseline model update
Escalation pathway	CMO alert trigger; mission control notification criteria

Critical design note: Anxiety vs. Depression
Anxiety response flowcharts operate on a 20-minute monitoring window — the condition is acute. Depression response flowcharts operate on a 48–72 hour monitoring window — the condition is chronic and slow-building. They require fundamentally different architectures. Depression also requires behavioural activation (nudging the crew member to do something) rather than just passive environmental calming.

■ PARTLY — see gap WHAT WE DID WELL — 5.3 D3 Response flowcharts — all nine stages; anxiety vs depression [AI protocols +50]

- Protocols exist for ALL SIX conditions (the brief asks for at least three), each with all nine stages: detection inputs, signal weighting, duration, triage, interventions per tier, notification & consent, monitoring window, resolution, escalation.
- Anxiety uses a 20-minute window; depression 48–72 h and behavioural ACTIVATION (greenhouse tending, Earth calls), not just calming — exactly the brief's critical note.
- The flowcharts are not just drawn — the engine actually runs them, so the board can watch a condition move through the stages live.

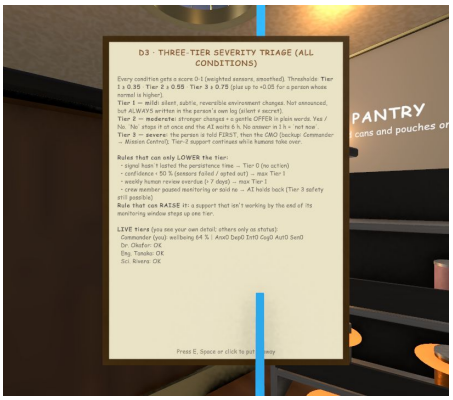
How we did it: D3 pages (12 in the sci-fi game) + the live engine; the triage page shows the rules that can only lower or raise a tier.

Gap / still to do: In the games the flowcharts are TEXT pages. For full marks, also show them as drawn diagrams on the slides (the FINAL deck already has the anxiety one).

Pictures from the simulations:



Realistic game (Moon_Base_Final): three-tier triage for all conditions



Sci-fi game (Lunar Habitat Commander): triage page

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6. Ethics and Hard Limits

In Phase 4 (Weeks 7–8), you address the hardest design questions. This section is worth 40 of 200 points — it is not an afterthought. The best designs are those where ethical reasoning shaped the technical architecture from the beginning, not ones where ethics were applied as a constraint at the end.

6.1 Four Core Ethical Tensions

Tension 1: Surveillance vs. privacy

The AI must watch crew members continuously to help them — but those crew members live in the habitat 24 hours a day, 7 days a week, for 18 months. There is no public/private boundary in the traditional sense.

Questions your design must answer:

- Where does monitoring stop? What spaces or times are excluded from sensor coverage?
- Who has access to individual sensor data? Can other crew members see it? Mission control?
- How is data stored, retained, and eventually deleted?
- What does crew consent look like when participation cannot be fully optional (a non-consenting crew member's physiological state affects mission safety)?

Tension 2: Autonomy vs. intervention

Every intervention the AI makes reduces crew autonomy to some degree. A habitat that constantly adjusts itself in response to the crew's emotional state could itself cause the autonomy deprivation it is designed to treat.

Questions your design must answer:

- If a crew member explicitly refuses an intervention, should the AI stop — even if the crew member's condition continues to deteriorate?
- How does the AI intervene for autonomy deprivation without its intervention itself feeling like an imposition?
- What is the distinction between the AI 'supporting' a choice and 'nudging' a choice?

Tension 3: Transparency vs. distraction

Informing crew members that the AI is adjusting their environment in response to their psychological state is both ethically correct and potentially harmful — knowing you are being monitored changes your behaviour, and constant notifications are themselves a stressor.

Questions your design must answer:

- When should the AI act silently vs. inform the crew member?
- How do you design transparency without it becoming intrusive?

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■ DONE ✓ WHAT WE DID WELL — 6. Ethics — Tension 1 (surveillance vs privacy) and Tension 2 (autonomy vs intervention) [Ethics +40]

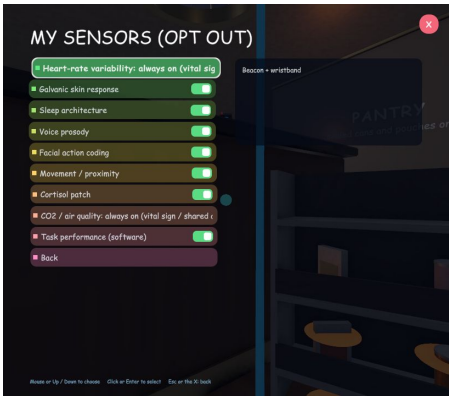
- Monitoring stops at the pod door: no cameras or mics in pods, hygiene or medical. Raw audio/video never leaves the room; only numbers do.
- Each person can opt out of any psychological sensor; heart-rate and CO2 stay on as the vital-sign safety floor (answers 'consent when participation can't be fully optional').
- A refused intervention STOPS and is logged; only a human may follow up. Supporting vs nudging is spelled out.

How we did it: optOut[] per person, OptOutAllowed[] keeps HRV and CO2 on; EthicsGuard blocks any AI action after a refusal for 6 h.

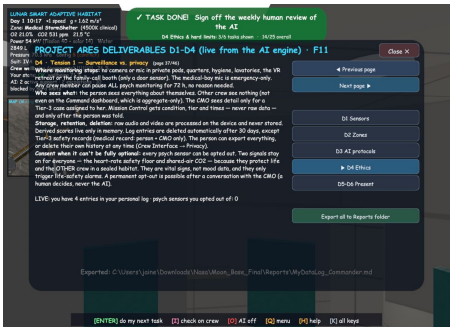
Pictures from the simulations:



Realistic game (Moon_Base_Final): privacy controls in the crew interface



Sci-fi game (Lunar Habitat Commander): opt out of any psych sensor



Realistic game (Moon_Base_Final): Tension 1 page

- Should crew members be able to see a log of all AI interventions made on their behalf?

Tension 4: Individual vs. crew

A crew member's private psychological data is simultaneously their personal information and a mission safety variable. If one crew member is at risk, other crew members' safety may depend on knowing — but that knowledge violates the first crew member's privacy.

Questions your design must answer:

- Can the AI act in a way that protects other crew members without revealing individual A's private psychological state?
- When does an individual's psychological crisis become a crew safety event?
- Who decides — the AI, the CMO, or mission control?

6.2 Hard Limits

Your ethics report must define clear hard limits — actions the AI is never permitted to take without explicit human authorization. Consider at minimum:

- Sharing individual crew member psychological data with other crew members without that individual's explicit consent
- Overriding a crew member's explicit refusal of an intervention (the system must back off and log)
- Taking any action that affects crew medication, medical protocols, or physical safety systems
- Alerting mission control about an individual's psychological state without informing that individual first
- Making any permanent change to habitat systems that cannot be quickly reversed by a crew member
- Operating for more than [your defined period] without a human system review

Phase 4 Deliverable: Ethics and Limits Report

800–1,200 words. Must address at least four ethical tensions with specific design solutions (not just acknowledgement of the tension). Must define hard limits with clear justification. Must specify what triggers human override of the AI system.

■ DONE ✓ WHAT WE DID WELL — Tensions 3–4 and the hard limits [Ethics +40]

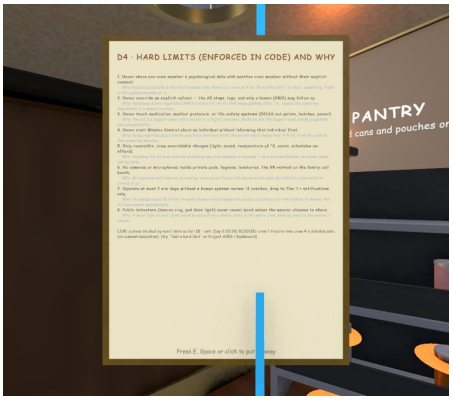
- Transparency without distraction: Tier 1 is silent but ALWAYS in the person's own log; anyone can turn on 'tell me about every silent change'.
- Individual vs crew: the AI acts on the environment and schedule, never on information — other crew are protected without revealing A's state; a crisis becomes a crew event only at Tier 3, decided by the CMO.
- Every hard limit in the brief is enforced IN CODE and can be tested live: no sharing data, no overriding a refusal, no medication or safety systems, no mission-control alert before the person is told, only reversible changes, and a weekly human review.

How we did it: EthicsGuard refuses blocked actions, counts them and shows them on the dashboard; 'Export my data' writes the full personal log.

Pictures from the simulations:



Realistic game (Moon_Base_Final): hard limits page



Sci-fi game (Lunar Habitat Commander): hard limits page



Realistic game (Moon_Base_Final): export everything the AI recorded about you



Sci-fi game (Lunar Habitat Commander): your data, exported

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7. Key Design Constraints

These constraints apply throughout all phases. Designs that ignore them will lose marks in the review board Q&A.

Mass and power budget	Every component sent to the Moon costs mass (approximately $1M/kg to the lunar surface) and draws power (every watt is power not available for life support). For each sensor and actuator you add, you must justify it in terms of mass, power draw, and mission-critical value. There are no free additions.
Crew autonomy is non-negotiable	No intervention may permanently override a crew member's explicit refusal. Your consent architecture must be built in from the beginning — not retrofitted. The system must have a clear and simple override mechanism accessible to every crew member at all times.
Minimal surveillance footprint	Raw sensor data must not be accessible to other crew members. The system observes individuals, not for individuals. Any crew member must be able to request a complete log of what the AI has recorded about them.
Failure mode planning	What happens when a sensor fails, the AI misclassifies a condition, or two signals directly contradict each other? Your design must gracefully degrade. Silently losing a detection capability is safer than triggering a false alarm. Define your failure modes explicitly.
No intervention that worsens autonomy deprivation	A system that constantly adjusts itself in response to the crew's psychological state can itself cause the autonomy deprivation it is designed to treat. Every intervention must be tested against the question: 'Does this increase or decrease the crew member's sense of agency?'
Human escalation must be fast	The AI is a support layer, not a replacement for human care. The escalation pathway from AI detection to CMO response must be designed for speed and clarity. Define maximum response times for each severity tier.

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■ DONE ✓ WHAT WE DID WELL — 7. Key design constraints

- Mass and power: every sensor has kg and W (13.1 kg, 110 W).
- Autonomy is non-negotiable: a simple override for every crew member at all times — pause 24/72 h, stop all support, or the AI OFF switch (O in the realistic game, F12 in the sci-fi game).
- Minimal surveillance: no raw data to other crew; anyone can request their complete log.
- Failure modes: graceful degradation (page 6). No intervention that worsens autonomy: 'Let them pick'.
- Human escalation is fast: Tier 1 < 1 min automatic, Tier 2 offer < 5 min, Tier 3 acknowledged < 15 min (acute) or 1 h.
- NEW — NIGHT WATCH (Human > AI): while you sleep, if anything goes wrong (an alarm, the AI breaking a bedtime rule, a Tier-3 emergency, a sensor failure) the AI STOPS itself, undoes its changes and wakes you; you are in charge until you let it back.

How we did it: AIAuthority.Disallow() rolls back every AI change; only a human can Allow() again.

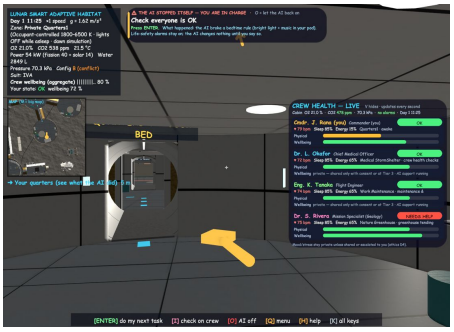
Pictures from the simulations:



Realistic game (Moon_Base_Final): the AI oversteps



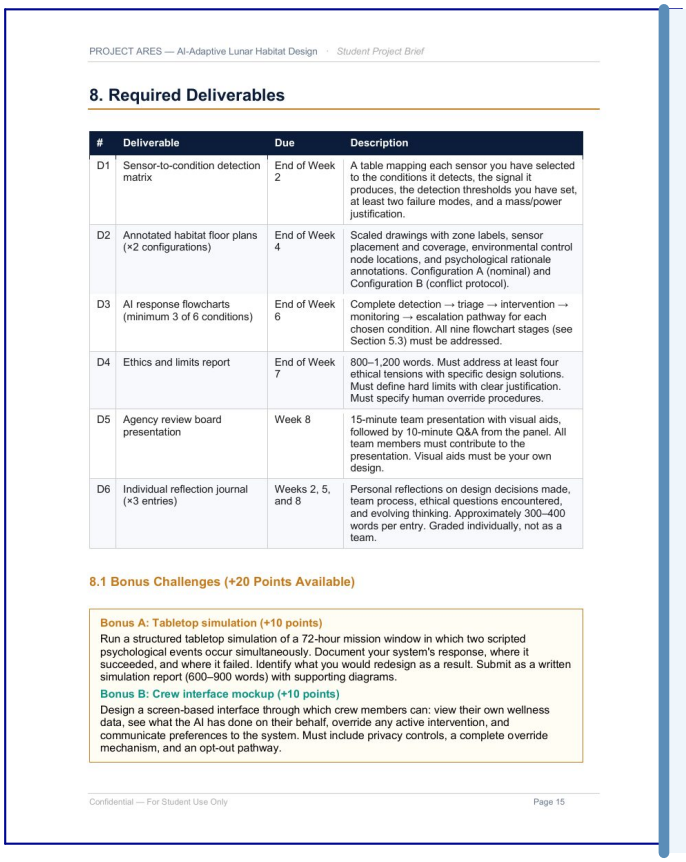
Realistic game (Moon_Base_Final): the AI OFF switch



Realistic game (Moon_Base_Final): NEW — night watch stopped the AI



Sci-fi game (Lunar Habitat Commander):
AI STOPPED — YOU ARE IN CHARGE



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8. Required Deliverables

#	Deliverable	Due	Description
D1	Sensor-to-condition detection matrix	End of Week 2	A table mapping each sensor you have selected to the conditions it detects, the signal it produces, the detection thresholds you have set, at least two failure modes, and a mass/power justification.
D2	Annotated habitat floor plans (×2 configurations)	End of Week 4	Scaled drawings with zone labels, sensor placement and coverage, environmental control node locations, and psychological rationale annotations. Configuration A (nominal) and Configuration B (conflict protocol).
D3	AI response flowcharts (minimum 3 of 6 conditions)	End of Week 6	Complete detection → triage → intervention → monitoring → escalation pathway for each chosen condition. All nine flowchart stages (see Section 5.3) must be addressed.
D4	Ethics and limits report	End of Week 7	800–1,200 words. Must address at least four ethical tensions with specific design solutions. Must define hard limits with clear justification. Must specify human override procedures.
D5	Agency review board presentation	Week 8	15-minute team presentation with visual aids, followed by 10-minute Q&A from the panel. All team members must contribute to the presentation. Visual aids must be your own design.
D6	Individual reflection journal (×3 entries)	Weeks 2, 5, and 8	Personal reflections on design decisions made, team process, ethical questions encountered, and evolving thinking. Approximately 300–400 words per entry. Graded individually, not as a team.

8.1 Bonus Challenges (+20 Points Available)

Bonus A: Tabletop simulation (+10 points)

Run a structured tabletop simulation of a 72-hour mission window in which two scripted psychological events occur simultaneously. Document your system's response, where it succeeded, and where it failed. Identify what you would redesign as a result. Submit as a written simulation report (600–900 words) with supporting diagrams.

Bonus B: Crew interface mockup (+10 points)

Design a screen-based interface through which crew members can: view their own wellness data, see what the AI has done on their behalf, override any active intervention, and communicate preferences to the system. Must include privacy controls, a complete override mechanism, and an opt-out pathway.

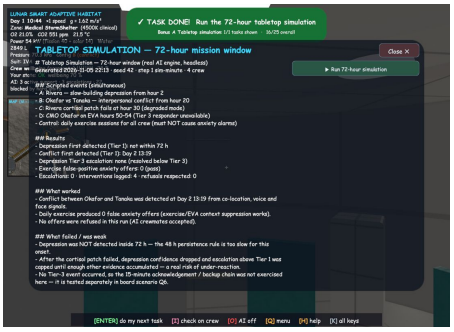
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■ DONE ✓ WHAT WE DID WELL — 8. Required deliverables D1–D6 + bonus challenges [Bonus A +10 · Bonus B +10]

- D1 matrix — in both games (page 5). D2 floor plans A/B — in both games (pages 7–8). D3 flowcharts — all six conditions (page 11). D4 ethics report — the D4 pages export as one report (Project ARES > Export all).
- D5 — a 15-minute run-of-show built INTO the simulator, and all six board questions run live (page 21). D6 — journal prompts are in the game; the journals themselves must be written by each of you.
- Bonus A — the 72-hour tabletop runs in the game and writes Reports/TabletopSimulation_Report.md. Bonus B — the crew interface has wellness, 'what the AI did for me', override, privacy, opt-out and preferences.

How we did it: Each deliverable is also a rubric task in the realistic game's checklist (J); the sci-fi game has a rubric self-check page.

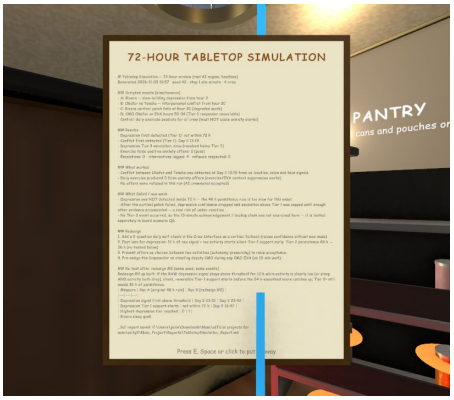
Pictures from the simulations:



Realistic game (Moon_Base_Final): Bonus A — 72-hour tabletop



Realistic game (Moon_Base_Final): Bonus B — crew interface



Sci-fi game (Lunar Habitat Commander): tabletop report



Sci-fi game (Lunar Habitat Commander): crew interface

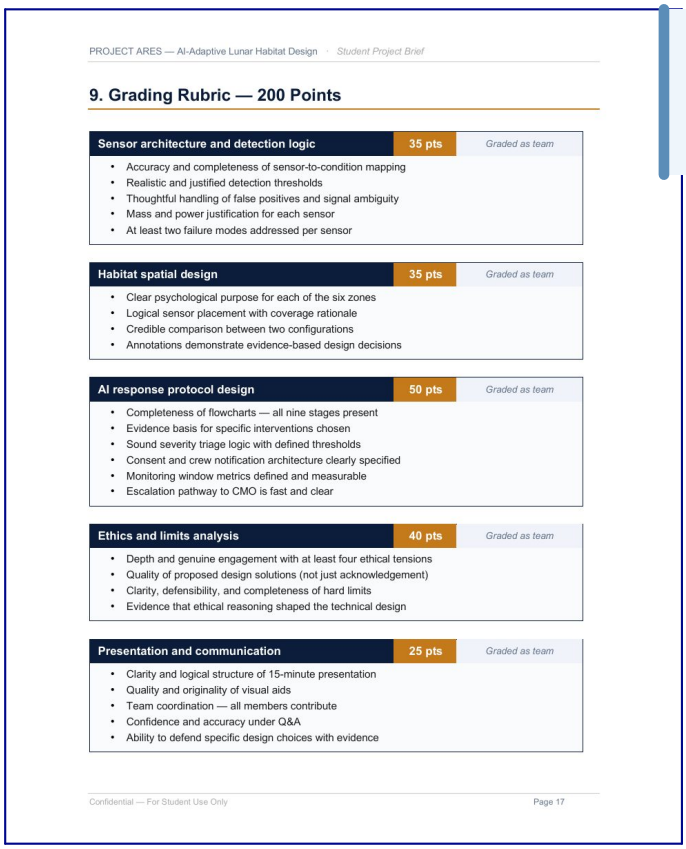
■ DONE ✓ WHAT WE DID WELL — Bonus challenges (continued) [Bonus A +10 · Bonus B +10]

- See page 15. Both bonus items are complete in both games.

Pictures from the simulations:



Realistic game (Moon_Base_Final): D5 run-of-show page



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9. Grading Rubric — 200 Points

Sensor architecture and detection logic	**35 pts**	*Graded as team*

- Accuracy and completeness of sensor-to-condition mapping
- Realistic and justified detection thresholds
- Thoughtful handling of false positives and signal ambiguity
- Mass and power justification for each sensor
- At least two failure modes addressed per sensor

Habitat spatial design	**35 pts**	*Graded as team*

- Clear psychological purpose for each of the six zones
- Logical sensor placement with coverage rationale
- Credible comparison between two configurations
- Annotations demonstrate evidence-based design decisions

AI response protocol design	**50 pts**	*Graded as team*

- Completeness of flowcharts — all nine stages present
- Evidence basis for specific interventions chosen
- Sound severity triage logic with defined thresholds
- Consent and crew notification architecture clearly specified
- Monitoring window metrics defined and measurable
- Escalation pathway to CMO is fast and clear

Ethics and limits analysis	**40 pts**	*Graded as team*

- Depth and genuine engagement with at least four ethical tensions
- Quality of proposed design solutions (not just acknowledgement)
- Clarity, defensibility, and completeness of hard limits
- Evidence that ethical reasoning shaped the technical design

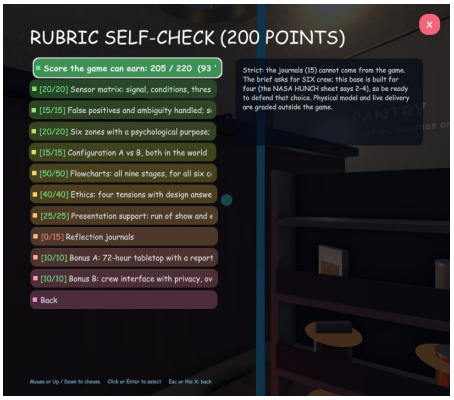
Presentation and communication	**25 pts**	*Graded as team*

- Clarity and logical structure of 15-minute presentation
- Quality and originality of visual aids
- Team coordination — all members contribute
- Confidence and accuracy under Q&A
- Ability to defend specific design choices with evidence

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■ SELF-GRADE WHAT WE DID WELL — 9. Grading rubric — strict self-grade

Pictures from the simulations:

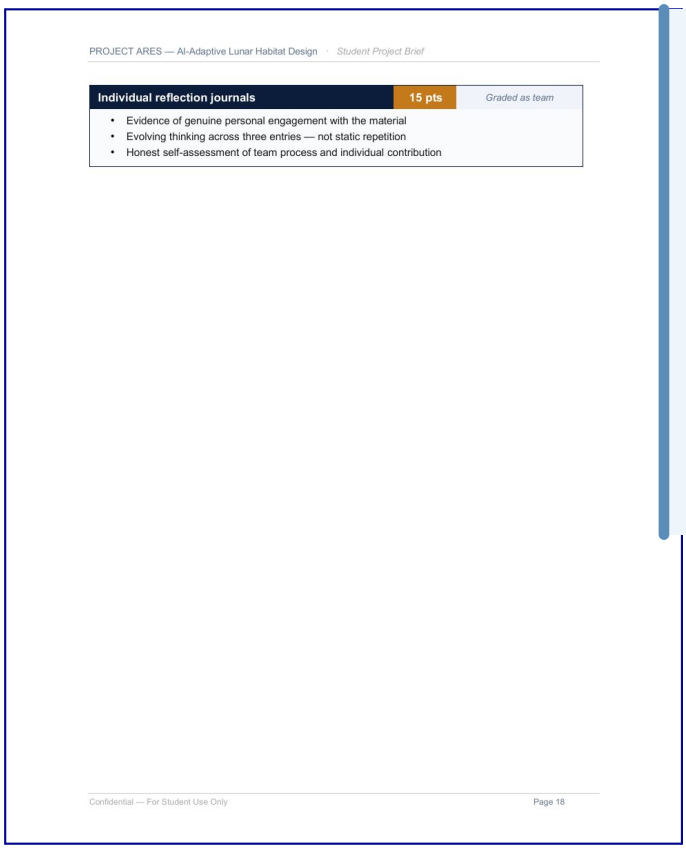


Sci-fi game (Lunar Habitat Commander): in-game rubric self-check

Self-grade, line by line (strict):

Rubric line	Points	Our grade	What we did / why points were taken off
Sensor architecture and detection logic	+35	**33/35**	Nine sensors, thresholds, 2+ failure modes each, mass/power, graceful degradation, live check-in. −2: thresholds are design values, not validated data.
Habitat spatial design	+35	**30/35**	Six zones with purpose in both bases, sensor placement, Config A/B in the world. −5: no scaled, dimensioned drawing inside the games; crew of 4 instead of the brief's 6.
AI response protocol design	+50	**46/50**	All six conditions × nine stages, live engine, tiers, consent, monitoring windows, fast CMO escalation. −4: flowcharts are text pages in the games (draw them on slides).

Rubric line	Points	Our grade	What we did / why points were taken off
Ethics and limits analysis	+40	**38/40**	Four tensions with design answers, hard limits enforced in code, override, night watch. −2: report must still be checked at 800–1,200 words when exported.
Presentation and communication	+25	**22/25**	Run-of-show and Q1–Q6 live in the simulator. −3: delivery, teamwork and Q&A are judged on the day.
Individual reflection journals	+15	**0/15**	Must be written by each student. Prompts are in the game.
Bonus A: tabletop simulation	+10	**10/10**	72-hour run with two simultaneous events + sensor failure + CMO on EVA; report saved.
Bonus B: crew interface mockup	+10	**10/10**	Wellness, what the AI did, override, privacy, opt-out, preferences.
Total (200 + 20 bonus)	**220**	**189/220**	**86% of everything; 92% of the points the games themselves can earn (journals are written by each student).**



■ SELF-GRADE WHAT WE DID WELL — Rubric (continued) — reflection journals

- The journals (15 pts) are graded on YOUR personal thinking, so the game cannot earn them. The prompts for all three entries are in the game (D6 page).
- Tip: use what you learned building the games — e.g. why you added the night watch, why HRV and CO2 can't be opted out — to show thinking that evolves.

Gap / still to do: Write the three 300–400-word entries yourselves (weeks 2, 5, 8).

Pictures from the simulations:



Realistic game (Moon_Base_Final): D6 journal prompts

10. Eight-Week Timeline

Weeks 1–2
Phase 1: Research and Sensor Architecture

- Review psychological literature on isolated and confined environments (ICE): Antarctic wintering-over data, ISS crew reports, submarine studies
- Research each sensor type: how it works, what it measures, what it costs in mass and power
- Build your sensor-to-condition detection matrix
- Identify at least two failure modes per sensor
- Reflection journal entry 1 due end of Week 2

Deliverable: D1: Sensor-to-condition matrix

Weeks 3–4
Phase 2: Habitat Spatial Design

- Design the physical layout of your habitat module — floor plan, zone functions, dimensions
- Determine sensor placement in each zone with coverage area calculations
- Create Configuration A (nominal) and Configuration B (conflict protocol)
- Write psychological rationale annotations for every major design decision
- Check each design decision against your key constraints list

Deliverable: D2: Annotated floor plans (×2 configurations)

Weeks 5–6
Phase 3: AI Response System Design

- Design complete response protocols for all six conditions
- Produce detailed flowcharts for your chosen three conditions (minimum)
- Design your consent and crew notification architecture
- Define monitoring window metrics for each condition
- Design your escalation pathway from AI detection to CMO to mission control
- Reflection journal entry 2 due end of Week 5

Deliverable: D3: AI response flowcharts (3+ conditions)

Weeks 7–8
Phase 4: Ethics, Hard Limits, and Presentation

- Write your ethics and limits report — treat this as a genuine design document, not an afterthought

■ DONE ✓ WHAT WE DID WELL — 10. Eight-week timeline (weeks 1–4)

- Phase 1 (sensors) and Phase 2 (spatial design) outputs are all inside the games and the slides; the research PDF covers the ICE literature (Antarctic, ISS, Mars-500, HI-SEAS).

Pictures from the simulations:



Realistic game (Moon_Base_Final): help screen — the easy way to play

- Test your design decisions against the four core tensions
- Define and justify your complete hard limits list
- Prepare your 15-minute agency review board presentation
- Rehearse Q&A scenarios, including simultaneous-condition challenges
- Reflection journal entry 3 due with final submission

Deliverable: D4: Ethics report · D5: Presentation · D6: Journals

■ DONE ✓ WHAT WE DID WELL — Timeline (weeks 5–8)

- Phase 3 (protocols) and Phase 4 (ethics, hard limits, presentation, Q&A rehearsal) are all playable; the board questions can be rehearsed live in the game.

Pictures from the simulations:



Realistic game (Moon_Base_Final): the simple menu



Realistic game (Moon_Base_Final): banner after the missions

11. The Agency Review Board

Your Week 8 presentation is evaluated by a panel acting as agency officials. They will probe your design choices — not just what you built, but why. Expect adversarial questions.

What the Board Will Ask

Q1: *You have a sensor detecting HRV drop and vocal tension simultaneously. Both reach Tier 1 thresholds. But the crew member just finished a hard EVA workout. How does your system avoid a false positive?*

Q2: *Crew member A shows strong depression indicators. At the same moment, crew members B and C are in a Tier 2 conflict episode. What does your system do simultaneously across both events?*

Q3: *A crew member discovers that the AI adjusted their lighting and played calming audio yesterday without telling them. They are furious. How does your design prevent this — and if it happens anyway, what is your recovery protocol?*

Q4: *Your cortisol sensor patch fails on Day 45. You won't have a replacement until resupply on Day 90. Walk us through your system's degraded-mode operation for those 45 days.*

Q5: *We notice your system has no mechanism for a crew member to permanently opt out of monitoring. Why not? Justify this design choice from both a safety and an ethics standpoint.*

Q6: *Your system escalates to the CMO. The CMO is currently on EVA and unreachable for 4 hours. What happens to your Tier 3 protocol during that window?*

Good luck. The Moon is waiting.

The best designs in this project will not be the ones with the most sensors or the most complex flowcharts. They will be the ones where every decision — technical, spatial, procedural, and ethical — is grounded in a genuine understanding of what it means to keep a human being psychologically healthy in the most isolated and demanding environment our species has ever attempted to inhabit.

■ DONE ✓ WHAT WE DID WELL — 11. The Agency Review Board — Q1 to Q6 [Presentation +25]

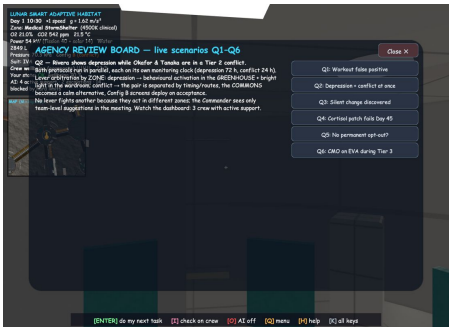
- Every board question is answered — and each one can be run LIVE in both games: the game sets the situation up on the real engine and then explains what the design does.
- Q1 workout false positive (exercise/EVA context lowers HRV, GSR and voice weights); Q2 depression + conflict at once (separate zones and clocks); Q3 silent change discovered (pre-mission consent, the log, apology and 'no Tier 1 for me'); Q4 cortisol patch fails Day 45 (degraded mode, self-check, weekly CMO chat); Q5 why no permanent opt-out (vital-sign floor, everything else can be paused); Q6 CMO on EVA during Tier 3 (Commander is backup, then mission control).

How we did it: Scenarios.Run(1..6). FIXED tonight: Q4 used to jump the game clock to Day 45; the clock is now put back after the scenario.

Pictures from the simulations:



Realistic game (Moon_Base_Final): Q1



Realistic game (Moon_Base_Final): Q2



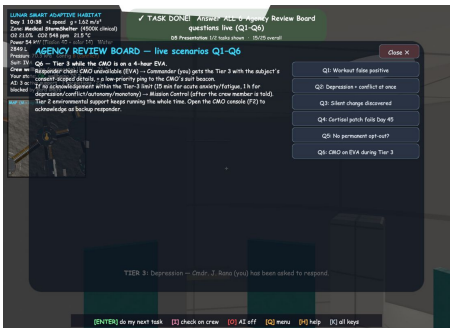
Realistic game (Moon_Base_Final): Q3



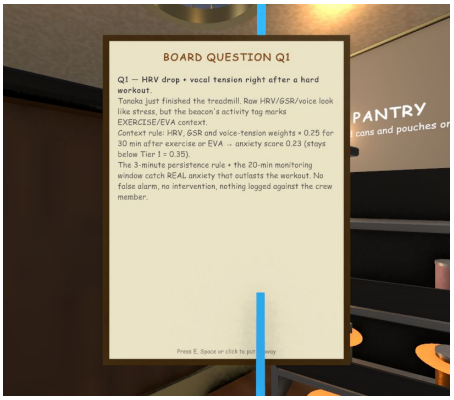
Realistic game (Moon_Base_Final): Q4



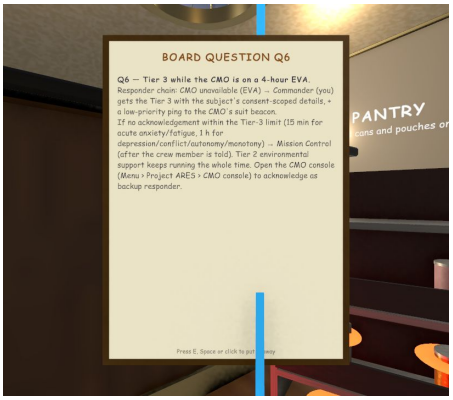
Realistic game (Moon_Base_Final): Q5



Realistic game (Moon_Base_Final): Q6



Sci-fi game (Lunar Habitat Commander): Q1



Sci-fi game (Lunar Habitat Commander): Q6